

IntelX Assurance Report

Report ID: REP-INTELX-6B4D3602 | Generated: 2026-09-05T06:21:55Z | Problem Statement 26228 | Ministry of Defence (MoD) / Indian Army (DGIS) | Policy: policy-2026.1

Overall Disposition: **QUARANTINE (risk score 100.0/100)**

Dataset	COMPROMISED
Model	VERIFIED
Inference Provenance	VERIFIED
Distribution Shift	ANOMALIES_DETECTED
Audit Chain	VALID

Findings (4)

ID	Asset	Type	Sev.	Conf.	Reason	Action
FINDING-DUP-001	ds_poisoned_02	near_duplicate_flooding	HIGH	0.96	Detected near-duplicate cluster #1 with 6 redundant samples.	REVIEW
FINDING-LBL-CONTRIBUTORBRAVO	ds_poisoned_02	systematic_mislabeling	HIGH	0.92	Contributor 'contributor_bravo' exhibits systematic label corruption (5/21 samples, 23.8% rate).	QUARANTINE
FINDING-OOD-001	ds_poisoned_02	ood_insertion	MEDIUM	0.91	Detected 1 out-of-distribution samples deviating from baseline terrain/sensor profile.	REVIEW
FINDING-TRIGGER-002	ds_poisoned_02	trigger_injection	HIGH	0.75	Corner-patch heuristic (no declared trigger reference supplied): 6 sample(s) exhibiting trigger/patch evidence.	REVIEW

Contributor / Source Risk Summary

Contributor	Total	Susp.	Dup.	Label	OOD	Trigger	Risk	Action
contributor_bravo	21	18	6	5	1	6	HIGH	QUARANTINE
contributor_alpha	10	0	0	0	0	0	LOW	ACCEPT
contributor_charlie	9	0	0	0	0	0	LOW	ACCEPT

Coverage Statement

Attack Class	Status	Validation Method
label_flipping	PARTIAL	Reference-model visual disagreement check (when reference_model_path is supplied) or declared-metadata cross-check (fallback)
systematic_mislabelling	PARTIAL	Batch-level error density aggregated from per-sample label disagreement evidence
near_duplicate_flooding	SUPPORTED	Perceptual differential hashing (dHash) & Hamming clustering
ood_insertion	SUPPORTED	Multivariate feature distance & Mahalanobis distribution scoring
trigger_backdoor_poisoning	SUPPORTED	Spatial frequency matched-filter correlation & real clean-vs-triggered inference comparison
unknown_trigger_reconstruction	PARTIAL	Neural Cleanse-style per-class trigger reconstruction + MAD anomaly-index test, executed via a real ONNX-to-PyTorch weight-transplanted differentiable model
model_substitution	SUPPORTED	Bitwise canonical SHA-256 digest comparison of the actual model file bytes
anomalous_model_behaviour	SUPPORTED	Real reference-vs-candidate inference comparison; ONNX initializer weight kurtosis/variance analysis (white-box only)
black_box_model_assessment	PARTIAL	Input/output behavioral probing only
hash_only_model_assessment	PARTIAL	File-level SHA-256 digest comparison only
inference_tampering	SUPPORTED	Cryptographic hash binding DAG & Ed25519 digital signature validation

Attack Class	Status	Validation Method
replay_detection	SUPPORTED	Cryptographic nonces, sequence numbers, and timestamp freshness windows
reordering_detection	SUPPORTED	Monotonic per-stream sequence-number tracking (rejects any record whose sequence_number does not exceed the last verified sequence)
audit_log_modification	SUPPORTED	SHA-256 hash chaining plus a per-entry Ed25519 signature, independently recomputed and verified end to end
distribution_shift	PARTIAL	Categorical divergence ratio, illumination delta, pixel-derived blur/contrast/resolution/compression-blockiness relative delta, and diagonal Frechet distance over real CNN embeddings (backend/drift/embedding_extractor.py), all vs. a declared reference
pytorch_torchscript_ingestion	SUPPORTED	torch.jit.load / torch.load(weights_only=True), real parameter counting, tested against real compiled/saved fixtures

Assumptions

- System operates in a strictly air-gapped, offline environment with no external network access.
- Reference baseline fingerprints and declared test batteries are stored in authenticated local storage.
- Ingested datasets adhere to valid COCO or YOLO annotation specifications.

Limitations

- Black-box model evaluation is limited to input/output behavioral probing (reference battery comparison); white-box weight/activation statistics are explicitly reported UNAVAILABLE rather than approximated.
- Backdoor/trigger detection matches against known trigger signatures (spatial frequency patch correlation, or whatever trigger pattern a caller explicitly probes for) for the primary check. Blind, gradient-based reconstruction of a never-specified trigger (Neural Cleanse) is also implemented, but only for models this system can bridge into a differentiable framework (currently its own detector-family ONNX graph) and only under WHITE_BOX access.
- Label-flip/mislabelling detection is only as strong as the reference model supplied for visual verification. Without one, it falls back to trusting contributor-declared metadata and will not catch errors on genuinely unannotated real-world data.
- PyTorch/TorchScript ingestion is validated against real scripted-module and state_dict fixtures, but ONNX remains the format exercised by the largest share of this system's test suite (backdoor/parameter analysis, behavioural batteries).
- Zero-day stealthy semantic triggers with extremely small perturbation norms, or a trigger style other than a spatially-local patch (Neural Cleanse's own core assumption), may still evade gradient-based reconstruction; the anomaly-index threshold was empirically calibrated against this system's own fixtures, not validated at scale against diverse real-world backdoors.
- Assurance evaluation provides empirical evidence and risk grading, but does not mathematically guarantee the total absence of unknown zero-day attacks.

Audit chain digest: 4e8e3878d60debea05800ba33b7a4f74aa2da8908e6e1cf715113eaea0e97c11